

252 Gbit/s, Polarization-multiplexed 128 QAM Fully Coherent Transmission over 10 km SMF and 40 m Wirelessly at 125 GHz

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Abstract We report a 252-Gbit/s polarization-division-multiplexed (PDM) fully coherent transmission in the 125 GHz band. By employing frequency-domain-equalization that compensates for both imperfect frequency responses and IQ imbalance, PDM 18 Gbaud 128 QAM signals are successfully transmitted over a 10 km SMF and 40 m wirelessly. ©2026 The Author(s)

Introduction

Recently, research and development aimed at 6G have progressed rapidly. In 6G networks, which require wireless peak rates exceeding 100 Gbit/s [1], optical mobile fronthaul (MFH) plays a critical role in efficiently transmitting high-capacity data signals to distributed antennas.

As such an MFH, fully coherent optical and wireless transmission has attracted significant attention [2,3]. A 582 Gbit/s transmission at a 124.5 GHz intermediate frequency (IF), has been demonstrated over 10 km of single-mode fibre (SMF) and over 3.1 m wirelessly by frequency-division- and polarization-division-multiplexed (PDM), probabilistically shaped (PS) 64 QAM [2]. However, this system employs heterodyne detection with a free-running local oscillator (LO). Therefore, digital carrier-phase compensation is required in the receiver.

By contrast, we have proposed an injection-locked heterodyne detection system, which enables carrier frequency conversion with low phase noise and a high signal-to-noise ratio (SNR) without digital carrier-phase compensation [4]. Using this system, we demonstrated a 108 Gbit/s, 18 Gbaud 64 QAM transmission over a 10 km SMF and 70 m wireless transmission in the 125 GHz band [4].

In this paper, we employ frequency-domain equalization (FDE) based on a 2×2 complex

transfer matrix [5]. The proposed FDE simultaneously compensates for waveform distortions arising from both the imperfect frequency responses of the electrical and optical components and the IQ imbalance in the analogue IQ mixer, including the I/Q gain mismatch and quadrature mismatch. This enables us to increase the modulation multiplicity to 128 QAM. Furthermore, by employing polarization-division multiplexing in both the optical fibre and the wireless link, we could demonstrate a 252 Gbit/s fully coherent transmission over a 10 km SMF and 40 m wirelessly in the 125 GHz band.

PDM fully coherent optical and wireless transmission system in 125 GHz band

Figure 1 shows the configuration of a PDM fully coherent optical and wireless transmission system operating in the 125 GHz band.

At a central office, the output of a $1.5 \mu\text{m}$ external cavity laser diode (ECLD) with an 8-kHz linewidth [6] was split into two branches. In one branch, the optical carrier was IQ-modulated with an 18 Gbaud, 128 QAM signal generated by an arbitrary waveform generator (AWG). In the AWG, a root-raised cosine filter with a roll-off factor of 0.05 was used to reduce the bandwidth of the data signal to 18.9 GHz ($18 \text{ GHz} \times 1.05$). In the other branch, we generated an optical pilot tone

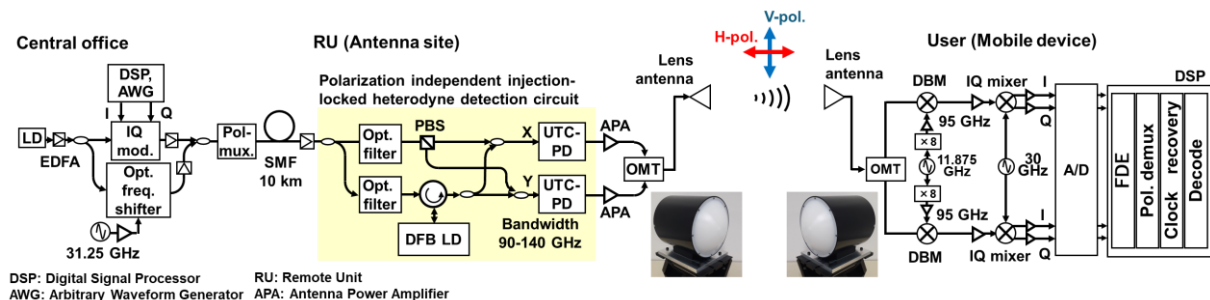


Fig.1: Experimental setup for PDM fully coherent transmission system in the 125 GHz band.

(PT) signal whose frequency was down-shifted by 125 GHz against the optical carrier by using an optical frequency shifter (OFS). The OFS consisted of a LiNbO₃ phase modulator driven by a 31.25 GHz sinusoidal wave and an optical filter with a bandwidth of 4 GHz. Here, a lower frequency 4th harmonic sideband of the phase-modulated signal was extracted using an optical filter as the PT signal. After combining the 128 QAM signal and the PT signal, 252 Gbit/s PDM signals (126 Gbit/s $\times$ 2 pol.) were generated using a polarization multiplexing circuit and transmitted over a 10 km SMF to a remote unit (RU) with a total optical power consisting of 0 dBm (-3 dBm per polarization) and -13 dBm for the PT. Figure 2 shows the optical spectrum of the PDM 18 Gbaud 128 QAM and the PT signals before the 10 km SMF transmission. The optical SNR with a 0.1 nm resolution bandwidth was as high as 44 dB.

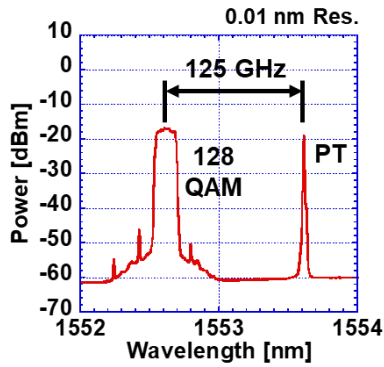


Fig.2: Optical spectrum of 18 Gbaud 128 QAM and PT signals before the 10 km SMF transmission.

At the RU, the PT signal was injected into a distributed feedback laser diode (DFB-LD), whose output was used as the phase-synchronized LO signal for the heterodyne detection [7]. The 128 QAM signal was separated into orthogonal X- and Y-polarization components with a polarization beam splitter (PBS) and heterodyne-detected with the LO using a uni-travelling carrier photodiode (UTC-PD) with a bandwidth of ~ 140 GHz to generate the 125 GHz band 128 QAM data signal. The input power into the UTC-PD was optimized at 9.3 dBm (Signal 0.3 dBm and LO 8.7 dBm). The 125 GHz band signal was amplified to an output power of -2 dBm and radiated via an orthomode transducer (OMT) as horizontal (H) and vertical (V) polarizations, enabling polarization-multiplexed wireless transmission. We used PDM lens antennas consisting of conical horn antennas and dielectric lenses with a diameter of 15.2 cm. The antenna gain and 3 dB beamwidth were 44.6 dBi and 1 degree, respectively.

On the user side, the two 125 GHz band

signals were polarization-separated by the OMT and down-converted to the 30 GHz band by using double-balanced mixers (DBM). Here, 95 GHz electrical LO signals were obtained by multiplying an 11.875 GHz sinusoidal signal 8 times. The 30 GHz band signals were then converted into baseband IQ signals using IQ mixers. The baseband signals were A/D-converted (80 GS/s, 33 GHz analogue bandwidth) and demodulated with digital signal processor (DSP).

In the DSP, FDE was employed to compensate for waveform distortions caused by both the imperfect frequency responses of the electrical and optical devices and IQ imbalance, followed by polarization demultiplexing [8]. In contrast, the conventional 99-tap finite impulse response (FIR) filter compensated only for distortions due to imperfect frequency response characteristics and was insufficient to compensate for IQ imbalance. These distortions can be represented by a 2×2 complex transfer matrix in the frequency domain as follows [5],

$$\begin{pmatrix} I_{rx}(\omega) \\ Q_{rx}(\omega) \end{pmatrix} = \begin{pmatrix} H_{II}(\omega) & H_{IQ}(\omega) \\ H_{QI}(\omega) & H_{QQ}(\omega) \end{pmatrix} \begin{pmatrix} I_{tx}(\omega) \\ Q_{tx}(\omega) \end{pmatrix} \quad (1)$$

where $I_{tx}(\omega)$ and $Q_{tx}(\omega)$ are the transmitted baseband IQ signals generated by the AWG, and $I_{rx}(\omega)$ and $Q_{rx}(\omega)$ are the received baseband IQ signals in the DSP. IQ imbalance introduces crosstalk between the I and Q channels, represented by $H_{IQ}(\omega)$ and $H_{QI}(\omega)$. The transfer matrix elements were estimated from training sequences in the frequency domain, and the distortions were compensated for by applying the inverse of the estimated transfer matrix to the received signals. In the FDE, a fast Fourier transform (FFT) was performed on a block of 4096 symbols (18204 samples), corresponding to a frequency resolution of 4.4 MHz. The FDE was applied independently to the V- and H-polarization data.

Experimental results

We evaluated the effectiveness of the proposed FDE scheme. Figures 3(a) and 3(b) show the constellations of an 18 Gbaud 128 QAM signal under a wireless back-to-back condition using the conventional 99-tap FIR filter-based equalization and the proposed FDE, respectively. With the FIR filter, the 128 QAM constellation does not converge due to insufficient compensation of waveform distortions. In contrast, with the FDE, 128 QAM signals were successfully demodulated with an error vector magnitude (EVM) of 3.4%. In wideband signal conversion using an analogue IQ mixer, the IQ imbalance is significantly increased, resulting in severe waveform

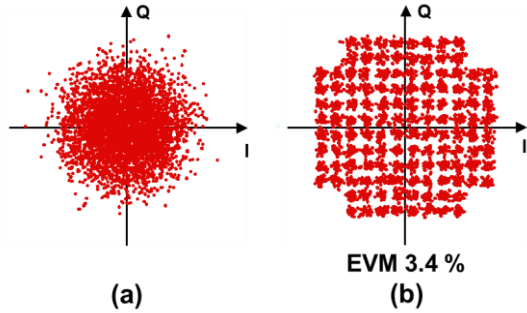


Fig. 3: Constellations of 128 QAM under a wireless back-to-back condition (a) with 99-tap FIR filter-based equalization and (b) with FDE.

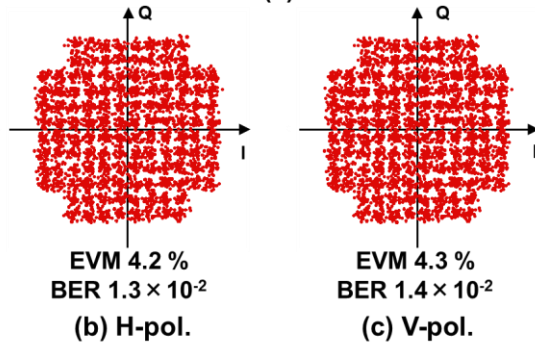
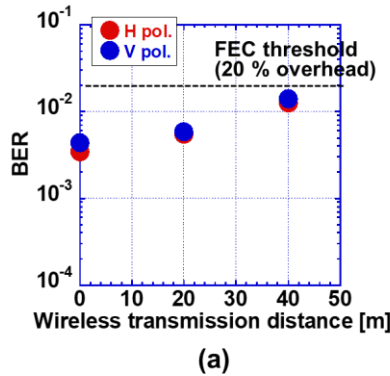


Fig. 4: (a) BER characteristics of PDM, 18 Gbaud 128 QAM signals as a function of wireless transmission distance, constellations of 128 QAM for (b) H-pol. and (c) V-pol. after a 40 m wireless transmission.

distortions. Therefore, FDE-based IQ imbalance compensation is indispensable for the accurate demodulation of high-order QAM signals in ultra-high speed fully coherent transmission systems.

Figure 4(a) shows the bit error rate (BER) characteristics of 18 Gbaud 128 QAM PDM signals as a function of wireless transmission distance. The degradation in BER beyond 20 m is attributed to the decrease in the received signal power, resulting in a lower SNR. After a 40 m wireless transmission, the BERs of the H- and V-pol. were 1.3×10^{-2} and 1.4×10^{-2} , respectively.

Figures 4(b) and 4(c) show the constellations of the 18 Gbaud 128 QAM signal for H- and V-pol. after a 40 m wireless transmission, respectively. The EVMs were 4.2 and 4.3 %, respectively. By applying a 20 % overhead forward error

correction (FEC) with a threshold of 2×10^{-2} , error-free transmission is expected to be achieved over a 40 m wireless link.

Conclusions

By employing FDE, which simultaneously compensates for waveform distortions arising from both imperfect frequency responses and IQ imbalance, a 252 Gbit/s, 128 QAM fully coherent transmission was demonstrated over 10 km of SMF and 40 m wirelessly, where we adopted an optical and wireless PDM technique. This optical and wireless transmission system is expected to enable ultra-high speed mobile fronthaul in future 6G networks.

Acknowledgements

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